Math 252Z [redacted] **Written Homework - 1.1 & 2.2**

1. Estimate the area under the graph $f(x) = 1 + x^2$ from $x = -1$ to $x = 2$ using six approximating rectangles and left endpoints. Sketch the graph of the rectangles. Is your estimate an underestimate or an overestimate?

2. Use the **limit of Riemann sum definition** of the definite integral to evaluate the following integral. (Note: No credit will be given if you use other method.)

1). $\displaystyle\int_0^2 (2 - x + x^2)\, dx$

2). $\int_1^2 x^3\,dx$

3. Evaluate the following integral by interpreting it in terms of areas.

1). $\int_{-2}^{2} \sqrt{4-x^2}\,dx$

2). $\int_0^{10} |x-4|\,dx$

4. Let $g(x) = \int_0^x f(t)\,dt$, where f is the function whose graph is shown. Evaluate $g(x)$ for $x = 1, 2, 3, 5,$ and 7.